

Swinburne University of Technology
School of Science, Computing and Emerging Technologies
Contribution Document

Subject Code: SWE30003

Unit Title: Software Architectures and Design

Assignment number and title: 3, Design Implementation

Due date: 05/08/2026

Tutorial Day and Time: Monday 16:00 – 19:00

Project Group: SG3_2

Tutor: Trung Pham

1. Time spent (hours) and contribution description

1.1 Fill in the time spent (hours) on each significant work item in the following table:

Work Item	Member 1 Name: Nguyễn Xuân Năng Mai	Member 2 Name: Đoàn Thiên Anh	Member 3 Name: Nguyễn Đức Trung
Read Assignment 3 brief & 120-point marking criteria	2	2	2
Review A2 design feedback & draft self-review matrix	3	2	2
Refine UML class diagram & GRASP Controller architecture	4	1	3
Prepare Sequence Diagram V1 (Browse, Search & Place Order)	3	0	0
Prepare Sequence Diagram V2 (Payment Processing)	0	3	0
Prepare Sequence Diagram V3 (Assign Vehicle & Driver)	0	0	3
Prepare Sequence Diagram V4 (Real-Time Shipment Tracking)	0	3	0
Design & create JSON seed data & bootstrap script (seed.ts)	0	0	4
Implement SmartFMController & Core Domain Layer	4	4	4
Implement V1 Order Flow & Home Dashboard UI	4	0	0
Implement V2 Payment Processing Flow & Receipt UI	0	4	0
Implement V3 Resource Assignment & Dispatch UI	0	0	4
Implement V4 Shipment Tracking & Driver 3-Tap UI	0	4	0

Implement Persistence Repositories (JsonFileRepository)	1	2	2
Joint integration testing & dependency chain verification	2	2	2
Capture 34 execution screenshots for Section 5 evidence	3	1	1
Write Report §1 (Design Revision) & §2 (Detailed Design)	4	0	0
Write Report §3 (Design Quality Reflection)	2	1	1
Write Report §4 & §5 (Implementation, Execution & Testing)	0	3	0
Write Report §6 (Architecture Styles) & §7 (Lessons Learnt)	0	0	4
Final report integration, ToC, Appendix A2 & packaging	2	0	0
Total hours	34	32	32

Note: Please adjust and add more rows, as necessary.

Description of contributions (each member provides a brief description of their contributions):

Member 1 Name: Nguyen Xuan Nang Mai:

- Owned Domain Classes: SmartFMController, Customer, Order, Invoice.
- Implemented Business Flow: Flow V1 (Browse Offerings → Search → Place Order) and Home Dashboard UI (app/orders/page.tsx, app/orders/[id]/page.tsx, app/page.tsx).
- Prepared & Drew Sequence Diagram: Seq-V1 (Browse, Search & Place Order).
- Report Sections Written:
 - Section 1: Summary of Design Revision (Table of revisions vs Assignment 2).
 - Section 2: Detailed Design (Class specifications for Control Layer & Ordering/Billing subsystem, GRASP Controller citation).
 - Section 3: Design Quality Reflection (Compiled team inputs for 5 Good, 5 Missing, 5 Flawed, 5 Ambiguous aspects).
 - Introduction & Conclusion sections.
- Project Setup & Canvas Submission: Set up project scaffolding, managed document integration, attached Appendix A (A2 Submission), and submitted final package on Canvas.

Member 2 Name: Doan Thien Anh:

- Owned Domain Classes: Shipment, TrackingUpdate, Payment, PaymentStrategy (Strategy Pattern & Factory).
- Implemented Business Flows:
 - Flow V2: Payment Processing & Receipt View (app/payment/page.tsx).
 - Flow V4: Real-Time Shipment Tracking & Driver 3-Tap Mobile Updates (app/tracking/page.tsx).
- Prepared & Drew Sequence Diagrams: Seq-V2 (Payment Processing) and Seq-V4 (Real-Time Tracking).
- Report Sections Written:
 - Section 4: Design-to-Code Mapping & Layer Architecture Alignment.
 - Section 5: Implementation, Execution Guide & Testing Evidence (Prerequisites, installation guide, 34 scenario execution screenshots, and input validation test matrix).
- Testing & Execution Evidence: Conducted full execution walkthroughs, tested setup guide on a non-team peer, and captured all 34 execution screenshots.

Member 3 Name: Nguyen Duc Trung:

- Owned Domain Classes: Branch, Vehicle, StaffMember, Driver.
- Implemented Business Flow: Flow V3 (Branch Resource Assignment - Vehicle & Driver Dispatch) (app/assignment/page.tsx).
- Prepared & Drew Sequence Diagram: Seq-V3 (Assign Vehicle & Driver to Shipment).
- Report Sections Written:
 - Section 6: Architectural Style(s) (Layered Architecture + Centralized Controller, Components, Connectors, Constraints, Justification).
 - Section 7: Lessons Learnt & Future Improvements (Compiled team technical reflections).
 - Assumptions & System Bootstrap writeup.
- Seed Data & Persistence: Designed and created all 9 JSON seed datasets (branches.json, vehicles.json, drivers.json, staff.json, etc.) and the bootstrap script (src/data/seed.ts). Implemented VehicleRepository and DriverRepository.

... ..

2. Evidence of collaboration / teamwork on each significant work item (eg, discussions, mutual revisions and reviews, etc.):

- Held weekly team syncs to review design decisions and align on GRASP Controller responsibilities.
- Conducted joint code reviews on pull requests across domain entities, service orchestrators, and persistence repositories.
- Executed a joint integration testing session to verify the end-to-end dependency chain across all 4 business flows (V1 → V2 → V3 → V4).
- Collaborated on compiling bullet points for Design Quality reflection (Section 3) and Lessons Learnt (Section 7).

3. Signed by all members:

ID Number	Name	Signature
<u>105549980</u>	<u>Nguyen Xuan Nang Mai</u>	<u>Mai</u>
<u>105506833</u>	<u>Đoàn Thiên Anh</u>	<u>Anh</u>
<u>105508266</u>	<u>Nguyễn Đức Trung</u>	<u>Trung</u>